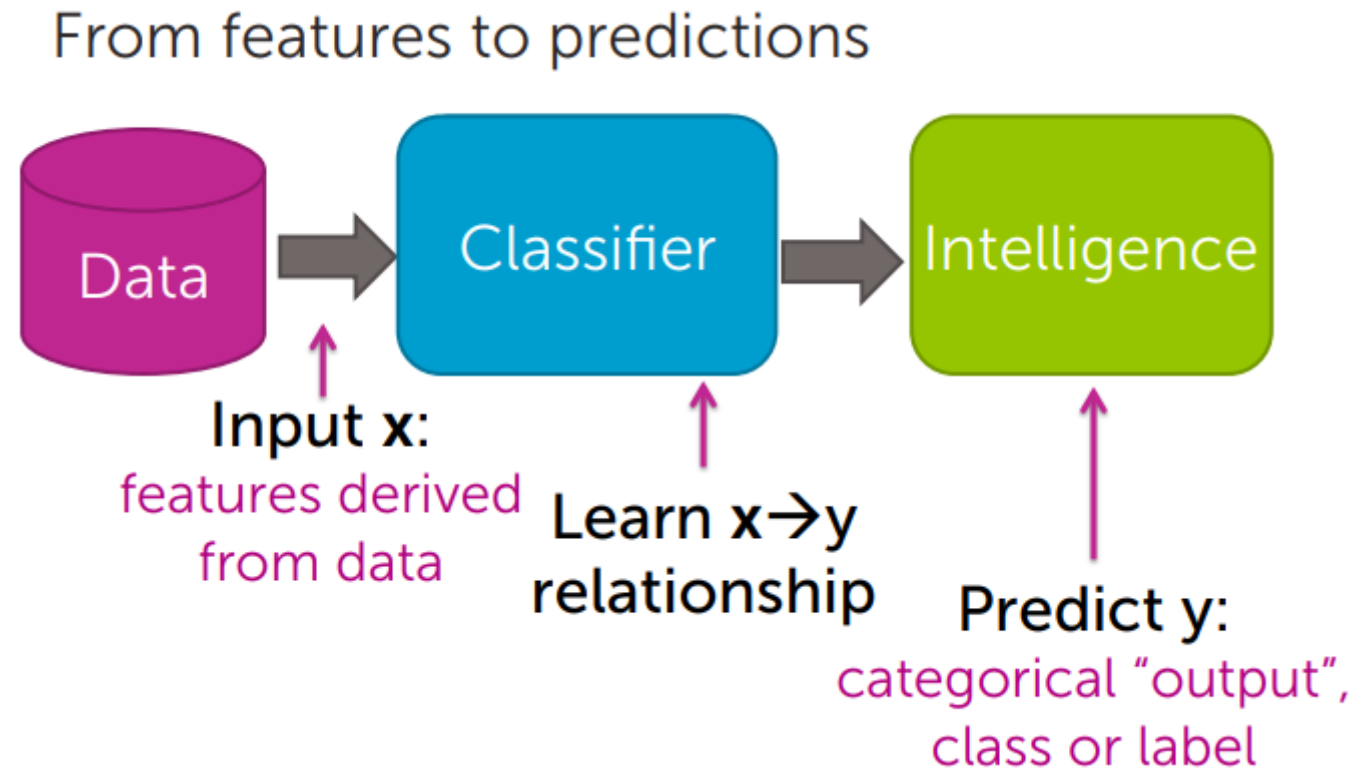


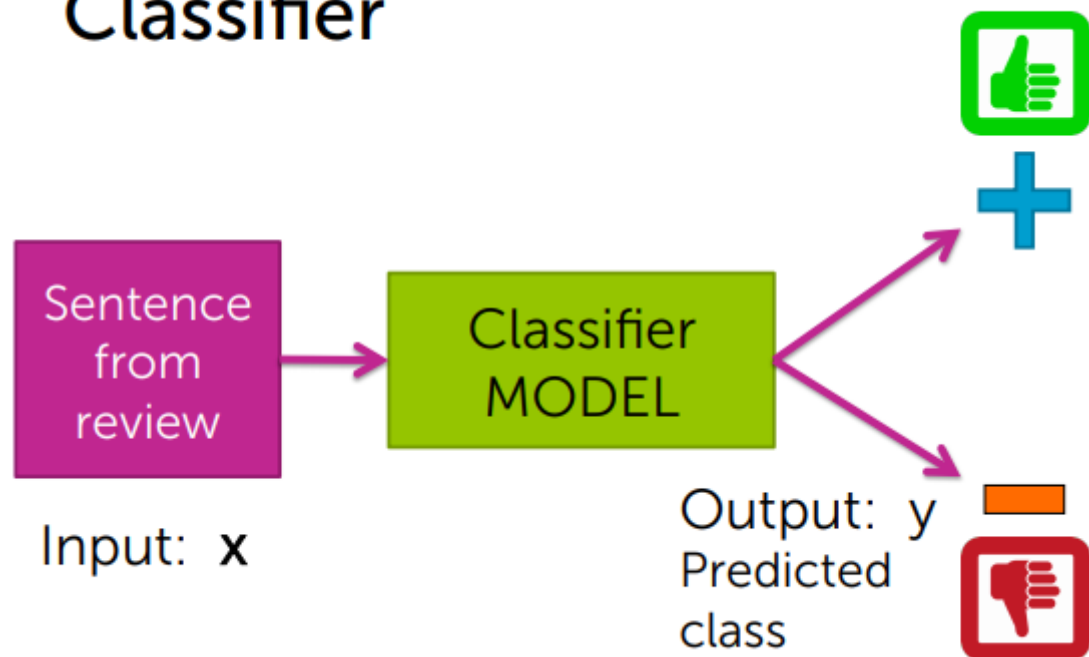
Modelos lineales de clasificación

Jhon Quiza

¿Qué hace un modelo de clasificación?

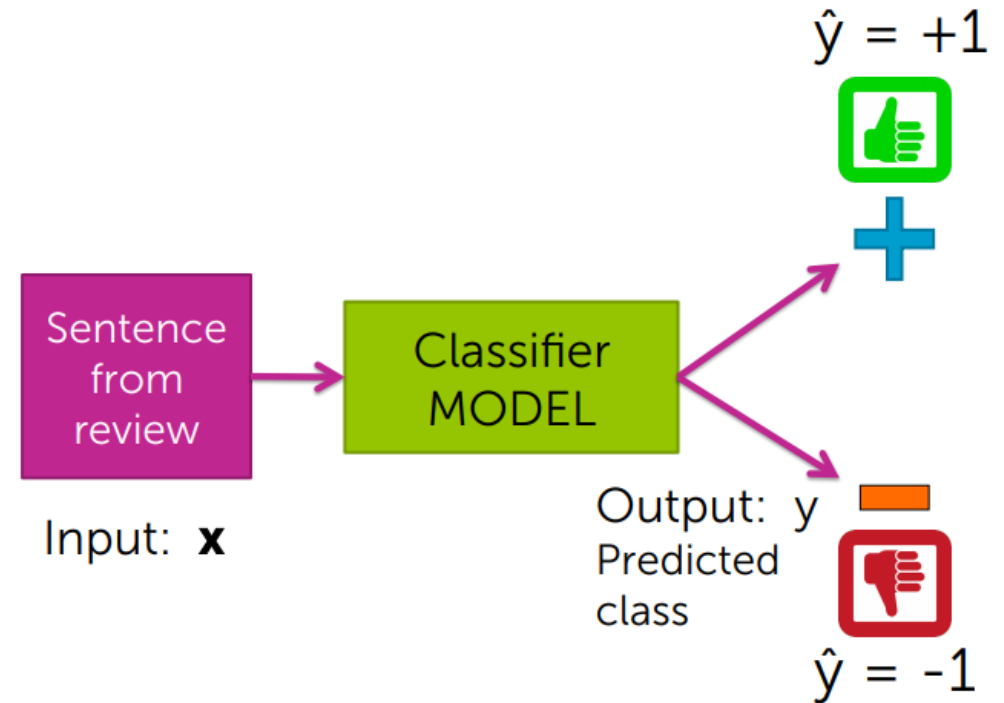
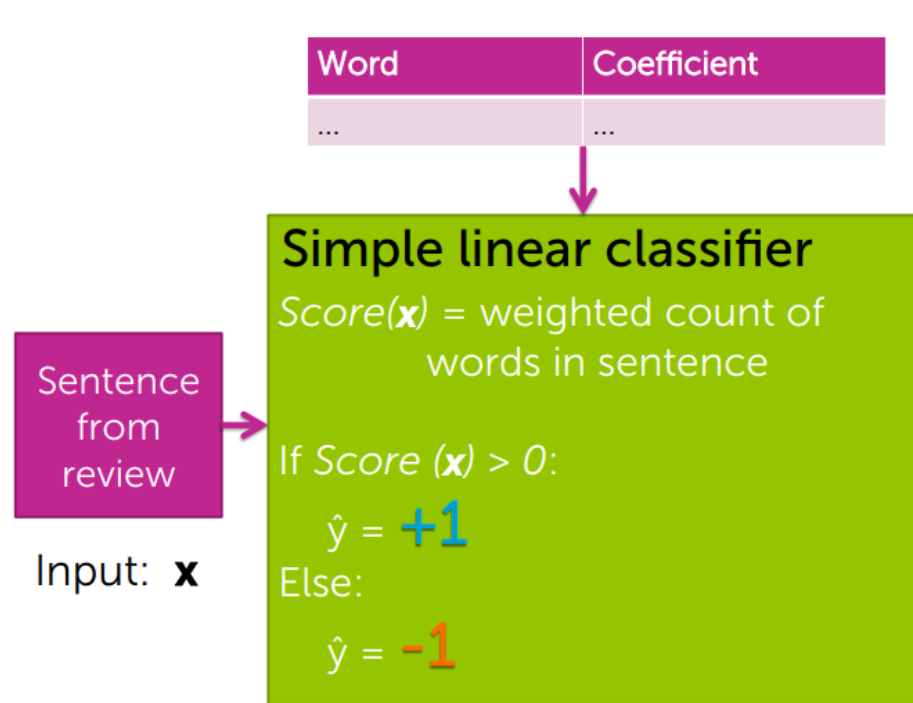


Classifier



Word	Coefficient
good	1.0
great	1.5
awesome	2.7
bad	-1.0
terrible	-2.1
awful	-3.3
restaurant, the, we, where, ...	0.0
...	...

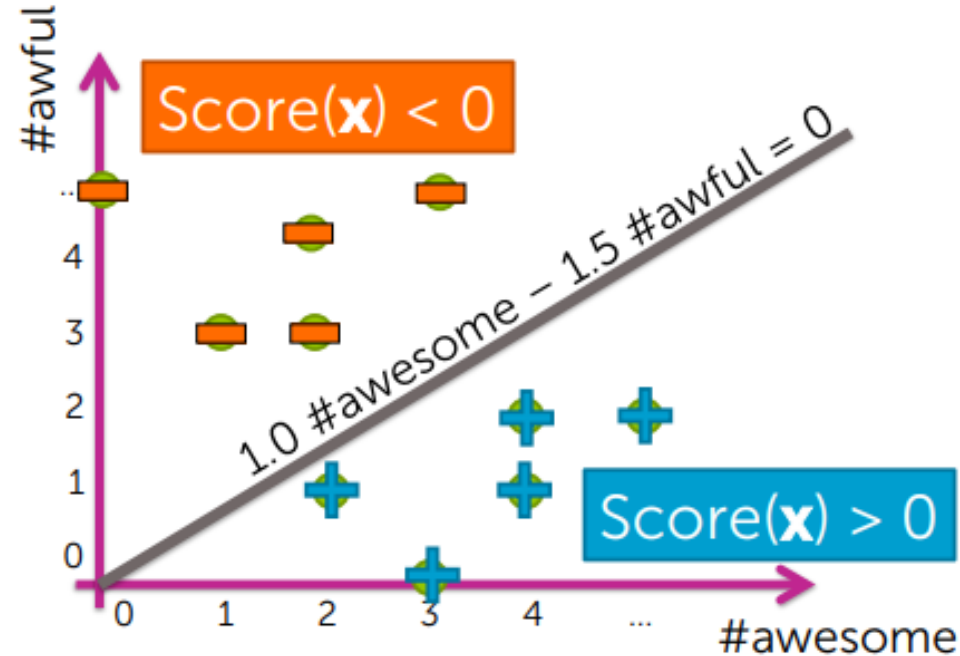
Función lineal: Score



Fronteras de decisión

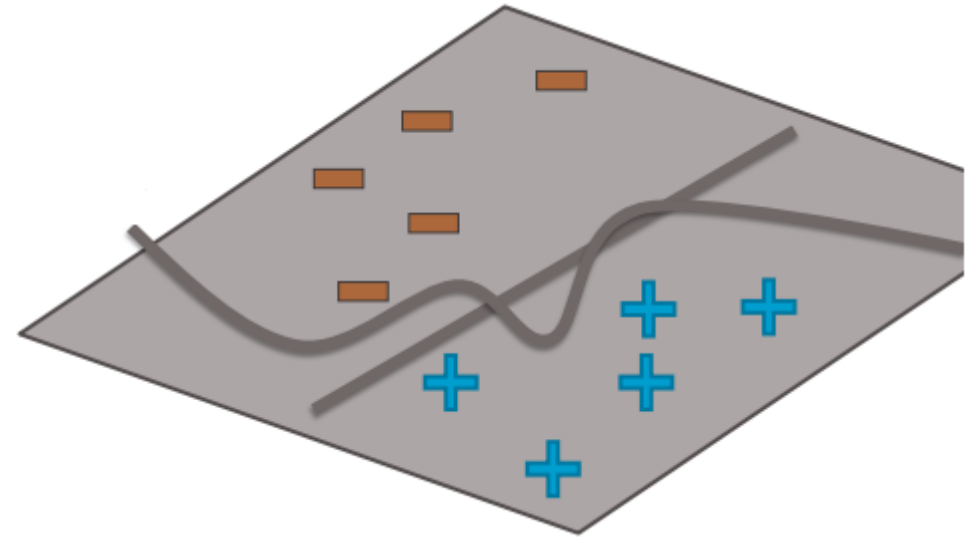
Word	Coefficient
#awesome	1.0
#awful	1.5

→ $\text{Score}(x) = 1.0 \text{ \#awesome} - 1.5 \text{ \#awful}$

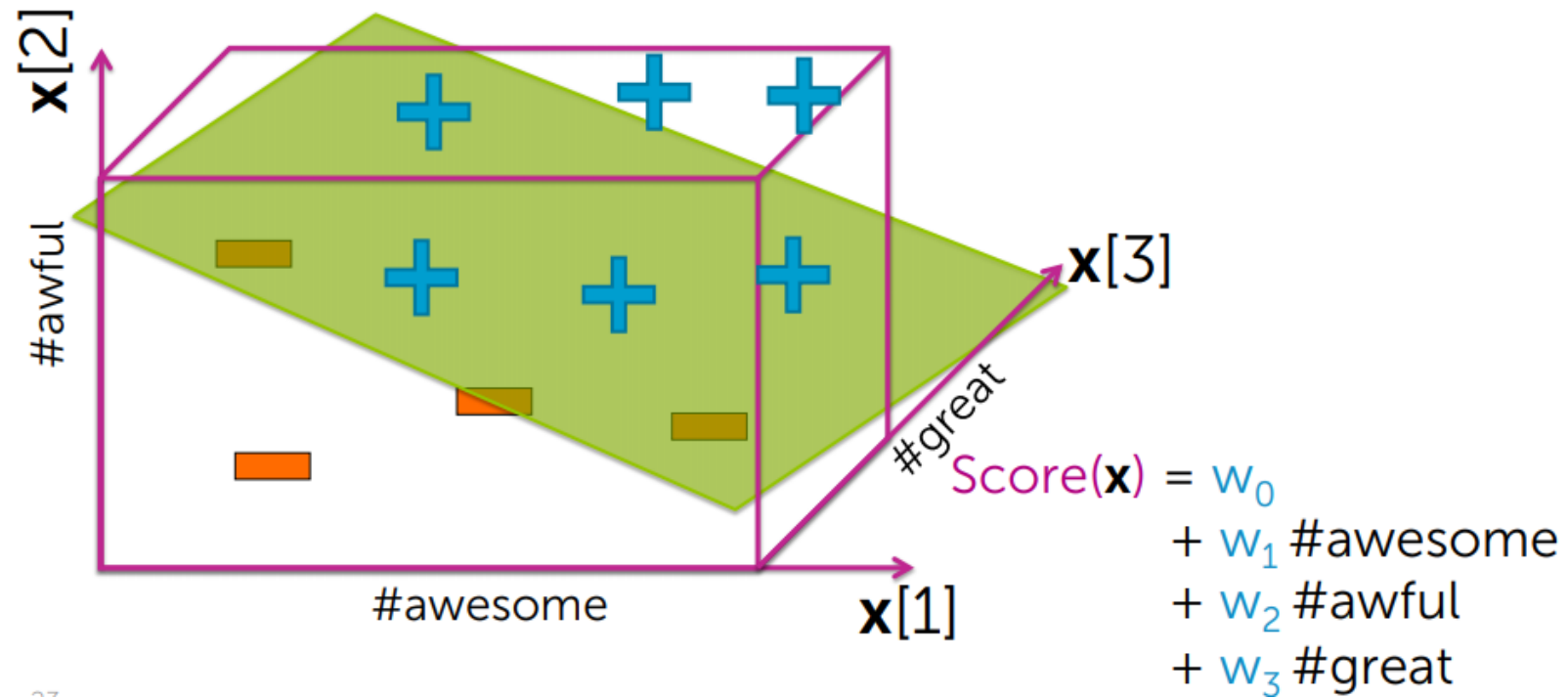


Fronteras de decisión con más de 2 características

- Para clasificadores lineales:
 - Cuando 2 coeficientes son diferentes de cero: línea.
 - Cuando 3 coeficientes son diferentes de cero: plano.
 - Cuando más de 3 coeficientes son diferentes de cero: hiperplano.
- Para clasificadores más generales las formas son más complicadas.



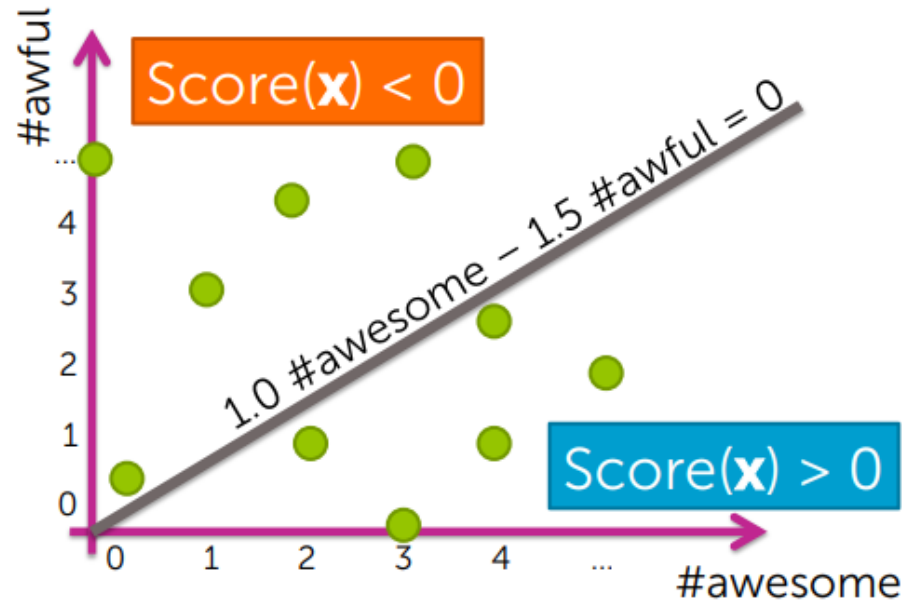
Modelo de clasificador lineal



Efecto de las fronteras de decisión

Input	Coefficient	Value
	w_0	0.0
#awesome	w_1	1.0
#awful	w_2	-1.5

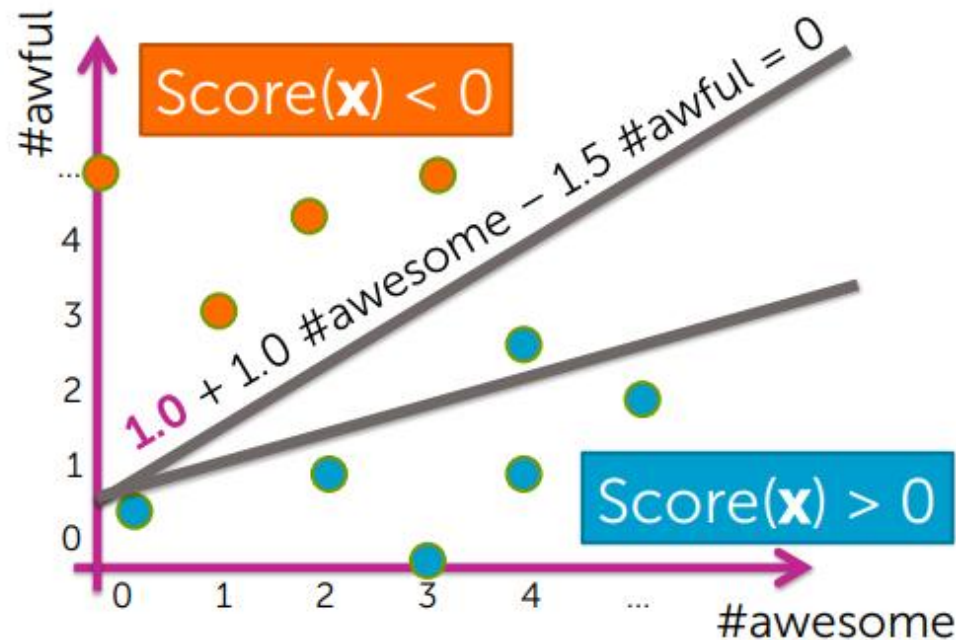
➔ $\text{Score}(x) = 1.0 \text{ \#awesome} - 1.5 \text{ \#awful}$



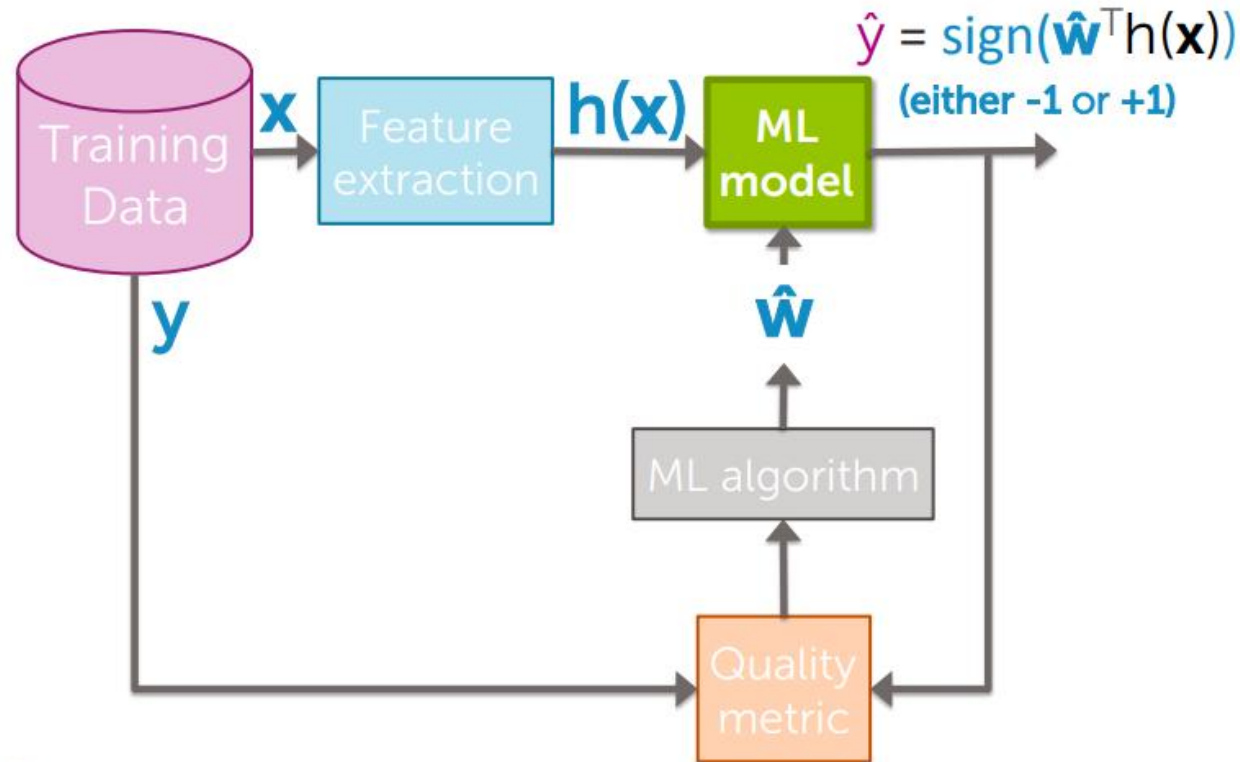
Efecto de las fronteras de decisión

Input	Coefficient	Value
	w_0	1.0
#awesome	w_1	1.0
#awful	w_2	-3.0

→ $\text{Score}(x) = 1.0 + 1.0 \text{ \#awesome} - 3.0 \text{ \#awful}$



Algoritmo para optimizar los coeficientes



Model: $\hat{y}_i = \text{sign}(\text{Score}(\mathbf{x}_i))$

$$\text{Score}(\mathbf{x}_i) = w_0 h_0(\mathbf{x}_i) + w_1 h_1(\mathbf{x}_i) + \dots + w_D h_D(\mathbf{x}_i)$$

$$= \sum_{j=0}^D w_j h_j(\mathbf{x}_i) = \mathbf{w}^T h(\mathbf{x}_i)$$

feature 1 = $h_0(\mathbf{x})$... e.g., 1

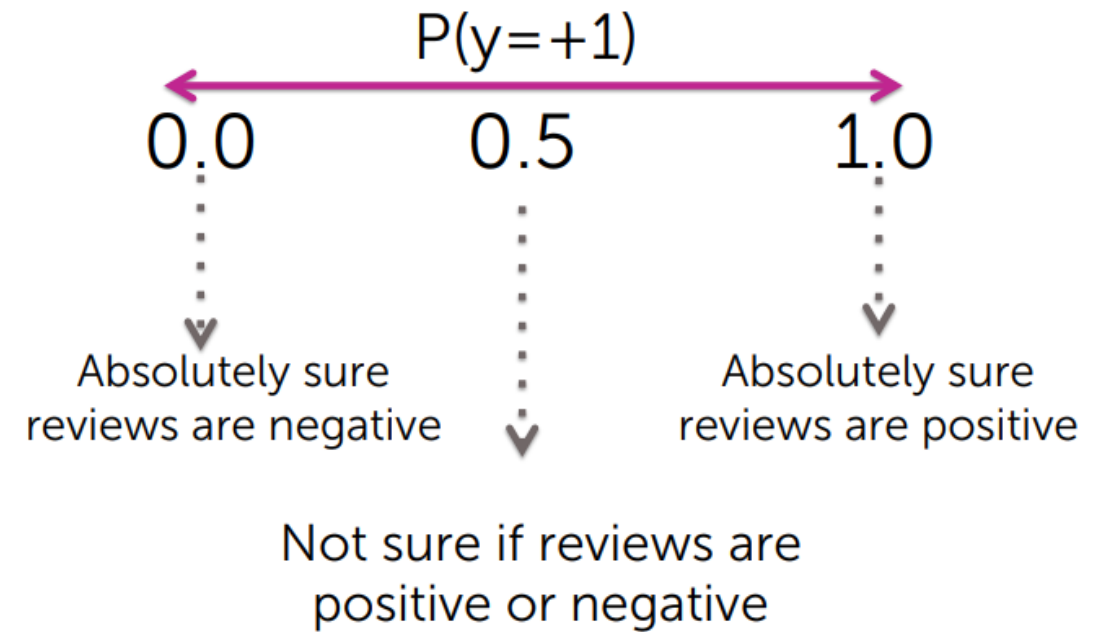
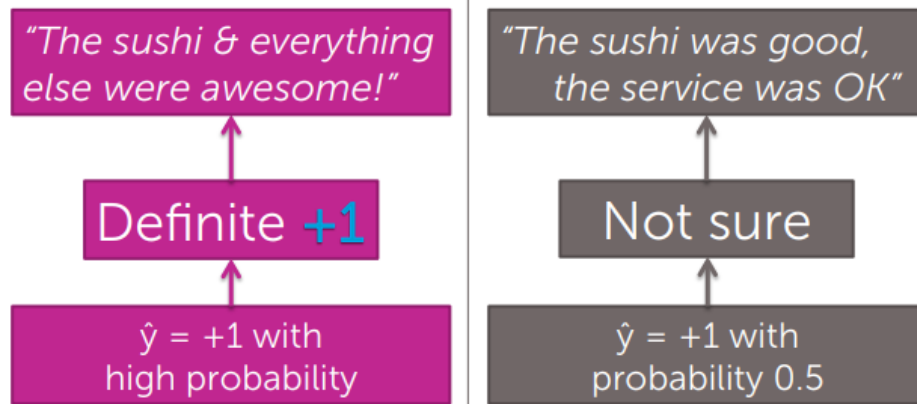
feature 2 = $h_1(\mathbf{x})$... e.g., $\mathbf{x}[1] = \text{\#awesome}$

feature 3 = $h_2(\mathbf{x})$... e.g., $\mathbf{x}[2] = \text{\#awful}$
or, $\log(\mathbf{x}[7]) \mathbf{x}[2] = \log(\text{\#bad}) \times \text{\#awful}$
or, $\text{tf-idf}(\text{"awful"})$

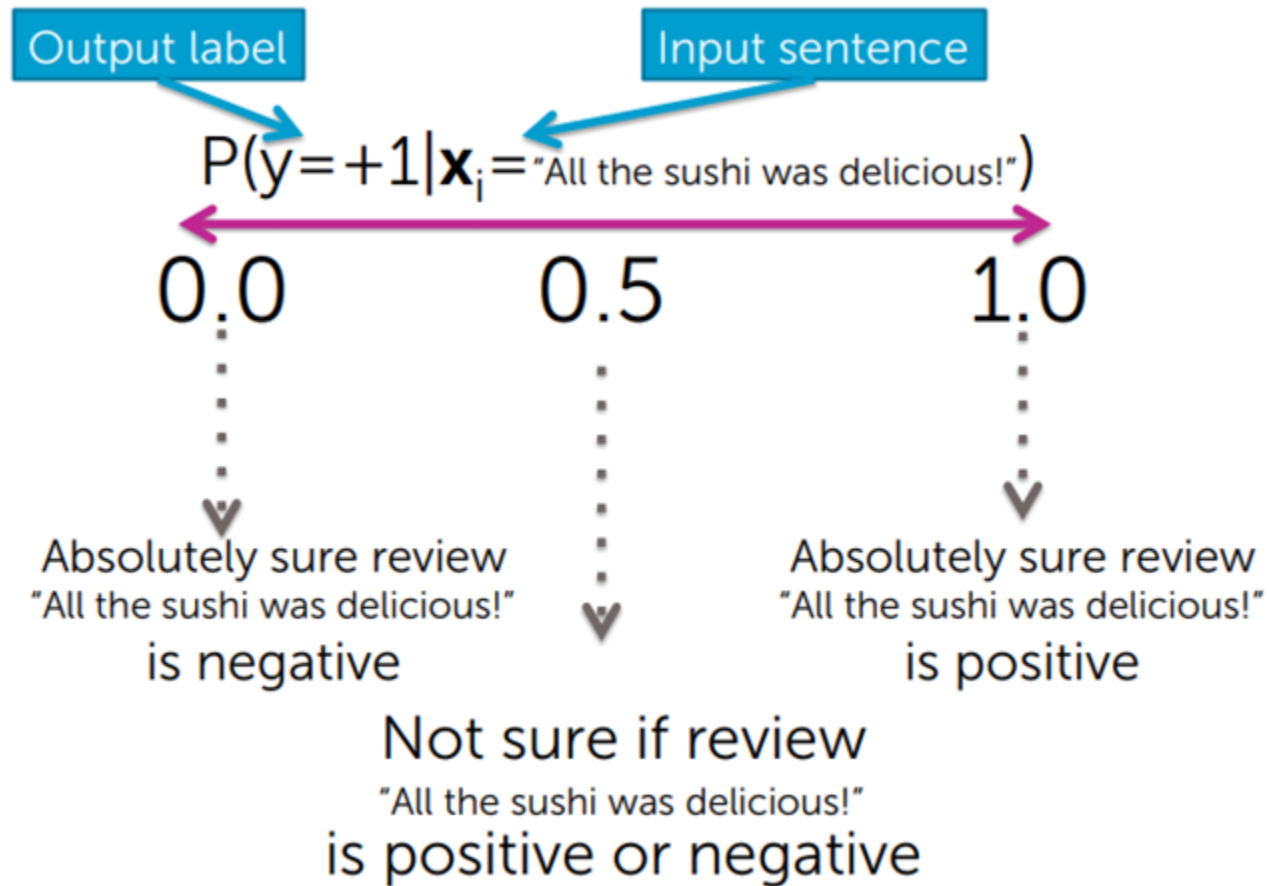
...

feature $D+1 = h_D(\mathbf{x})$... some other function of $\mathbf{x}[1], \dots, \mathbf{x}[d]$

Función de costo – probabilidad condicional



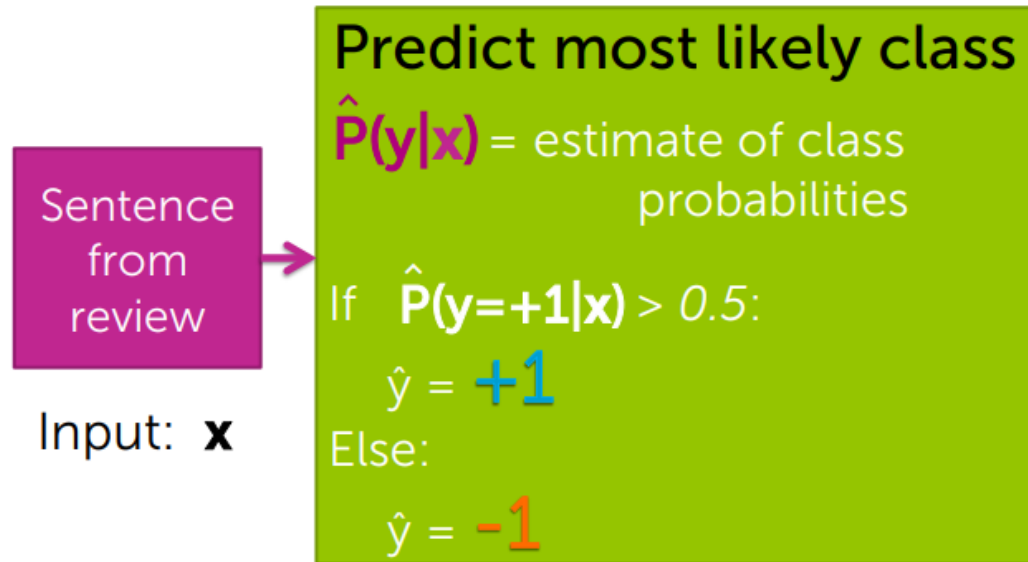
Probabilidad condicional



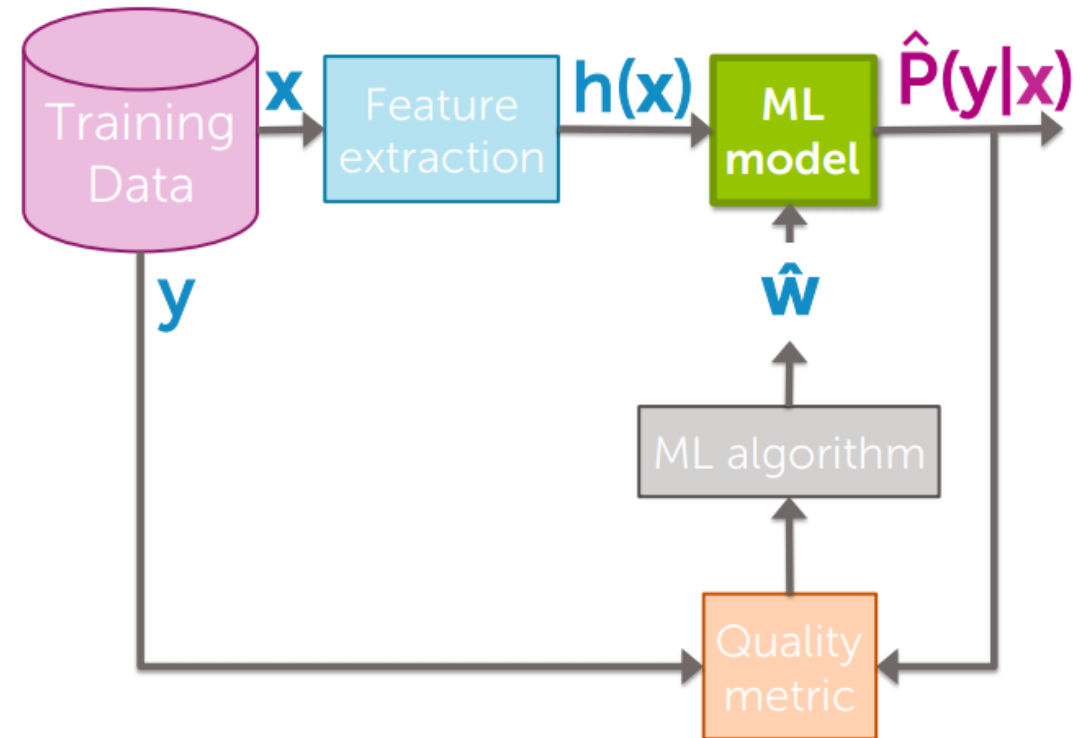
Probabilidades
condicionales siempre
están entre 0 y 1.

Probabilidades
condicionales suman 1
sobre y , pero no sobre x .

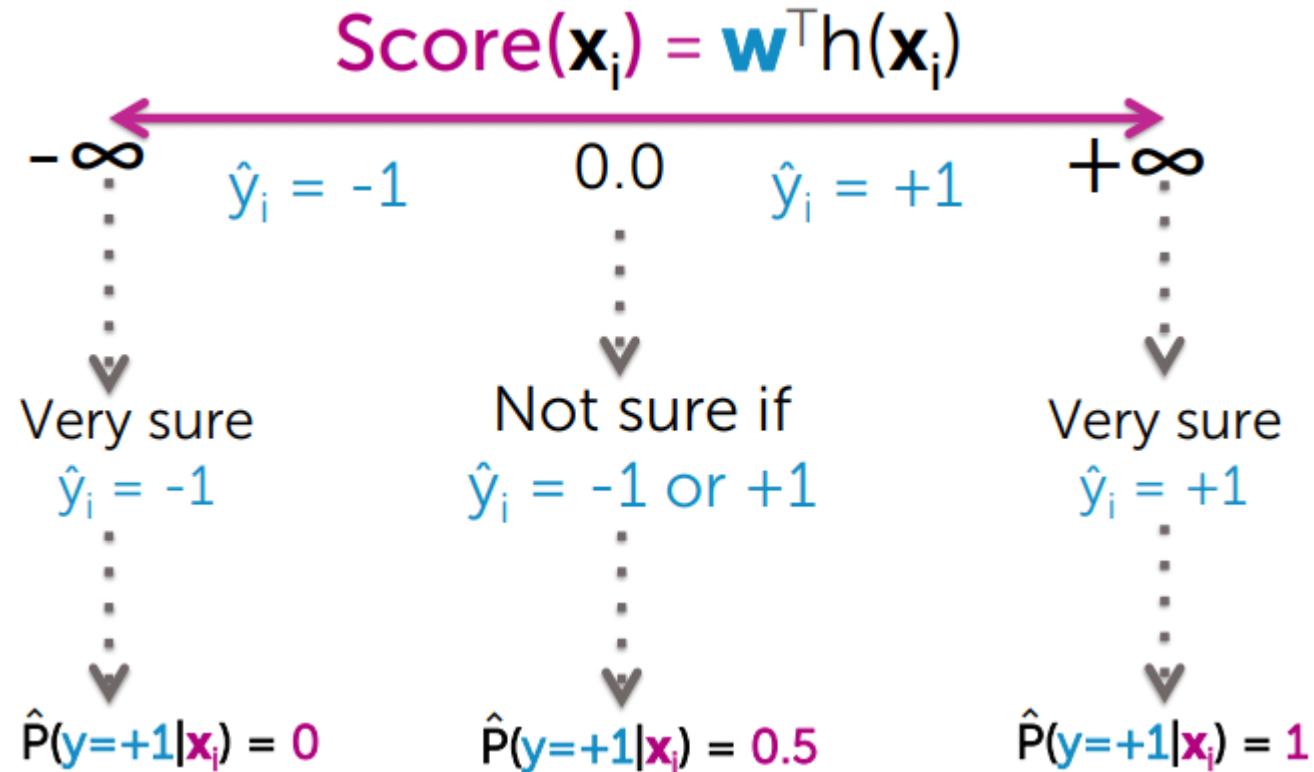
Modelo de clasificación probabilístico



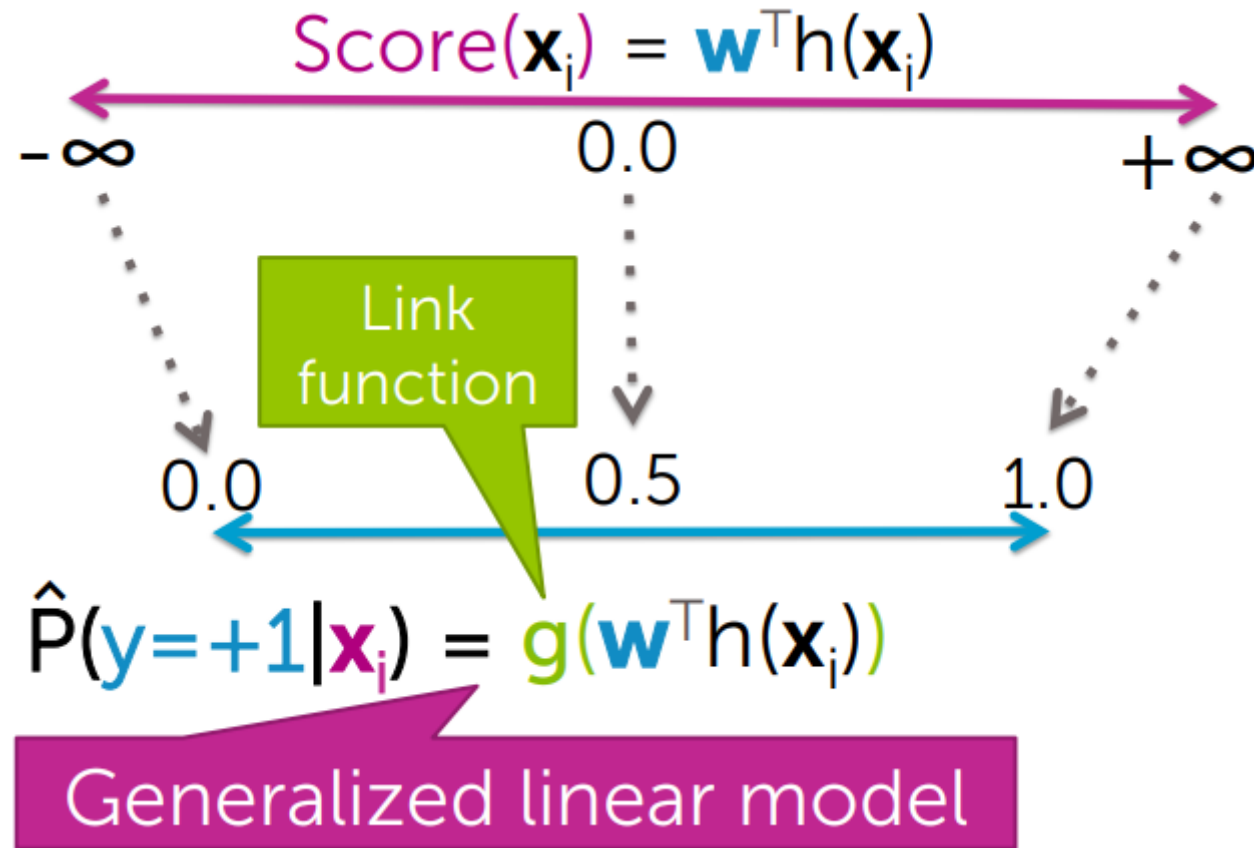
- Estimating $\hat{P}(y|\mathbf{x})$ improves **interpretability**:
 - Predict $\hat{y} = +1$ **and** tell me how sure you are



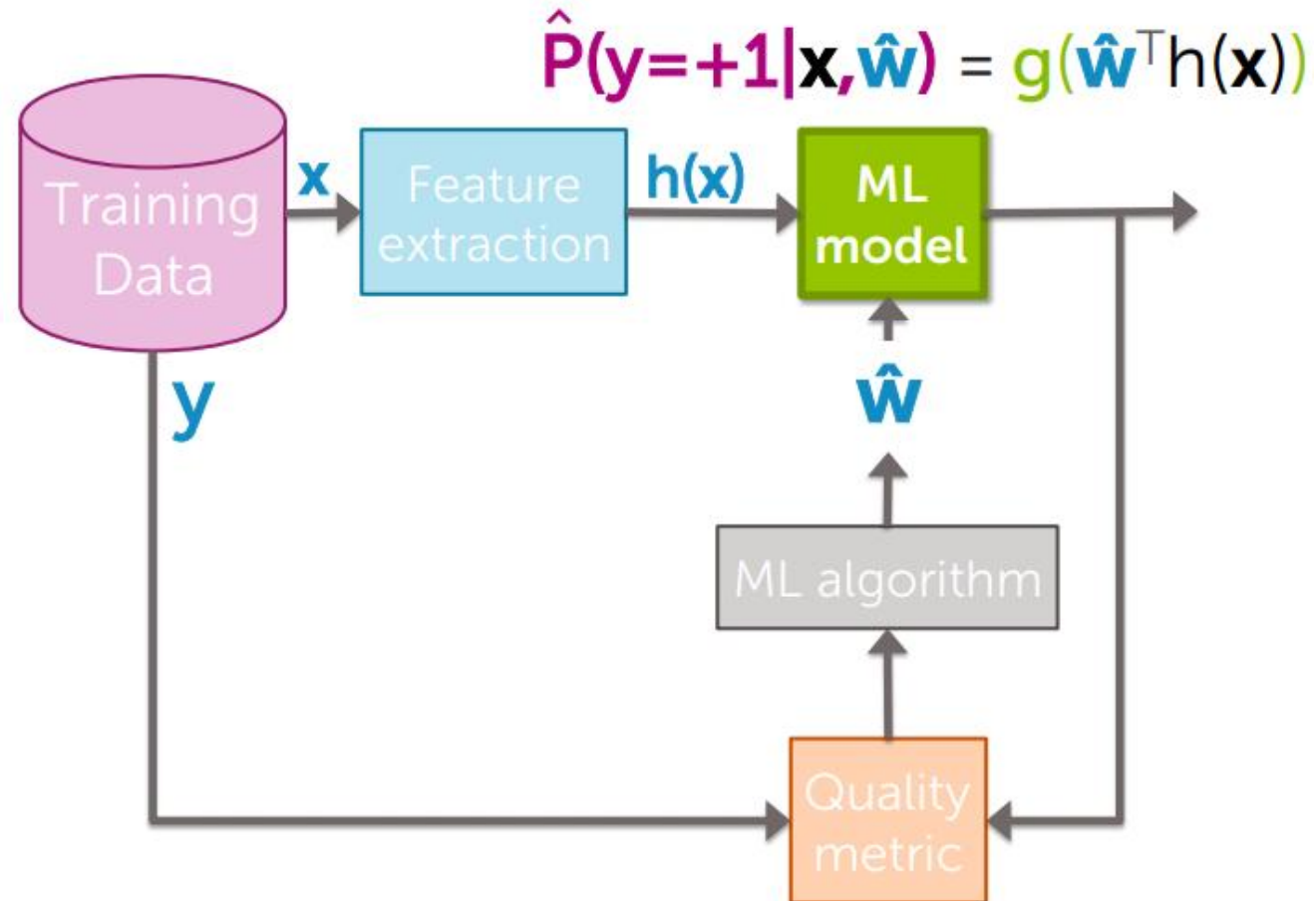
¿Cómo relacionar scores y probabilidades?



Función de enlace



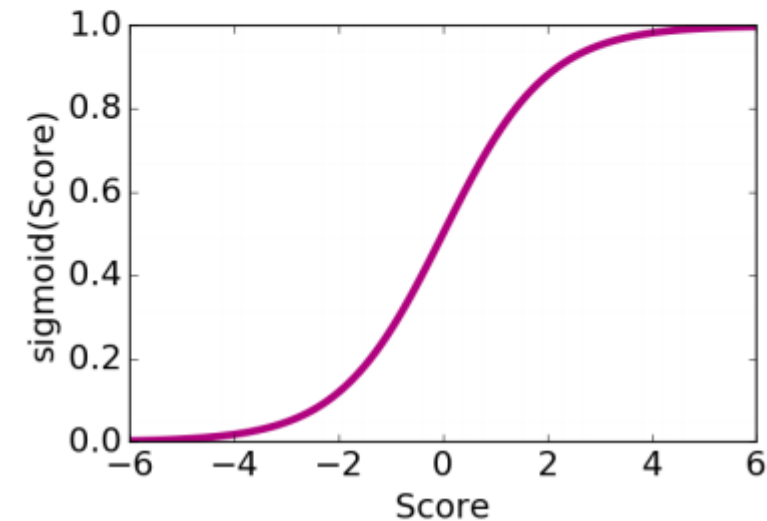
Modelo de clasificador lineal



Función logística o sigmoide como función de enlace

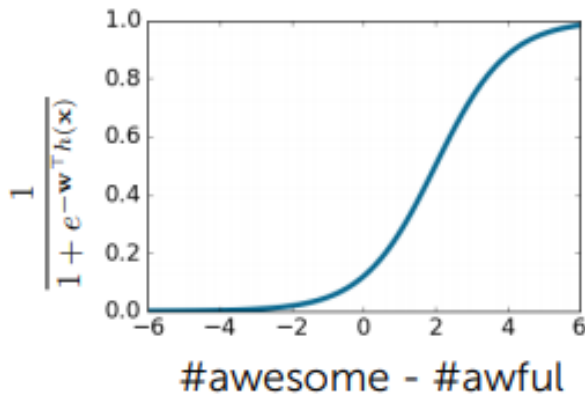
$$\text{sigmoid}(\text{Score}) = \frac{1}{1 + e^{-\text{Score}}}$$

Score	$-\infty$	-2	0.0	+2	$+\infty$
sigmoid(Score)					

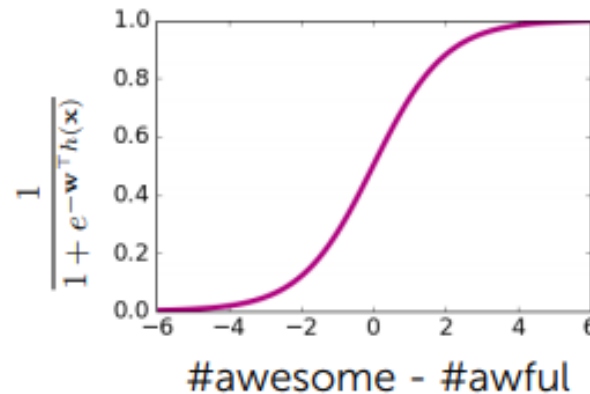


Efecto de los coeficientes en la función logística

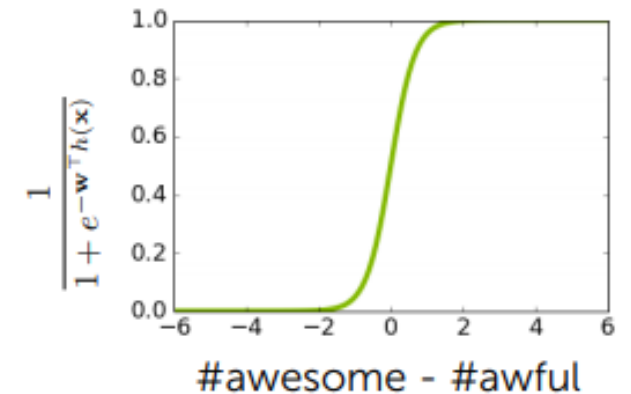
w_0	-2
$w_{\text{\#awesome}}$	+1
$w_{\text{\#awful}}$	-1



w_0	0
$w_{\text{\#awesome}}$	+1
$w_{\text{\#awful}}$	-1



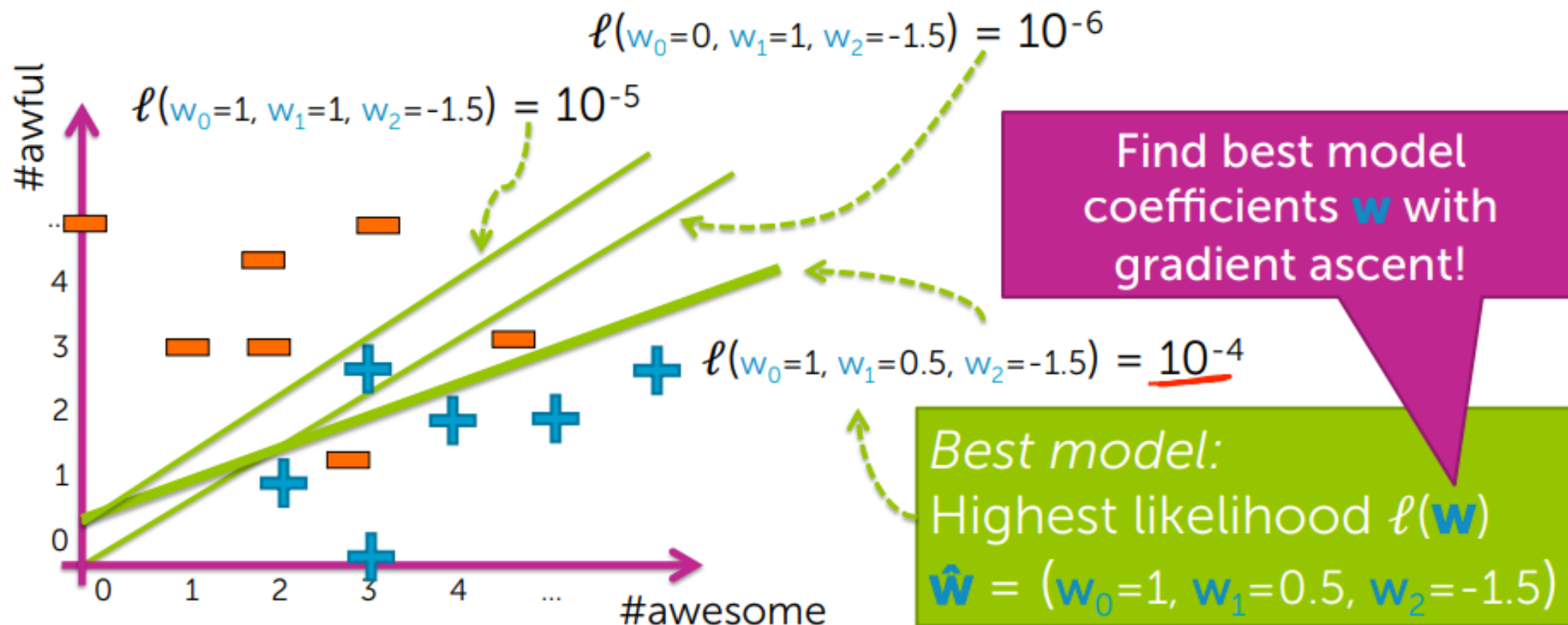
w_0	0
$w_{\text{\#awesome}}$	+3
$w_{\text{\#awful}}$	-3



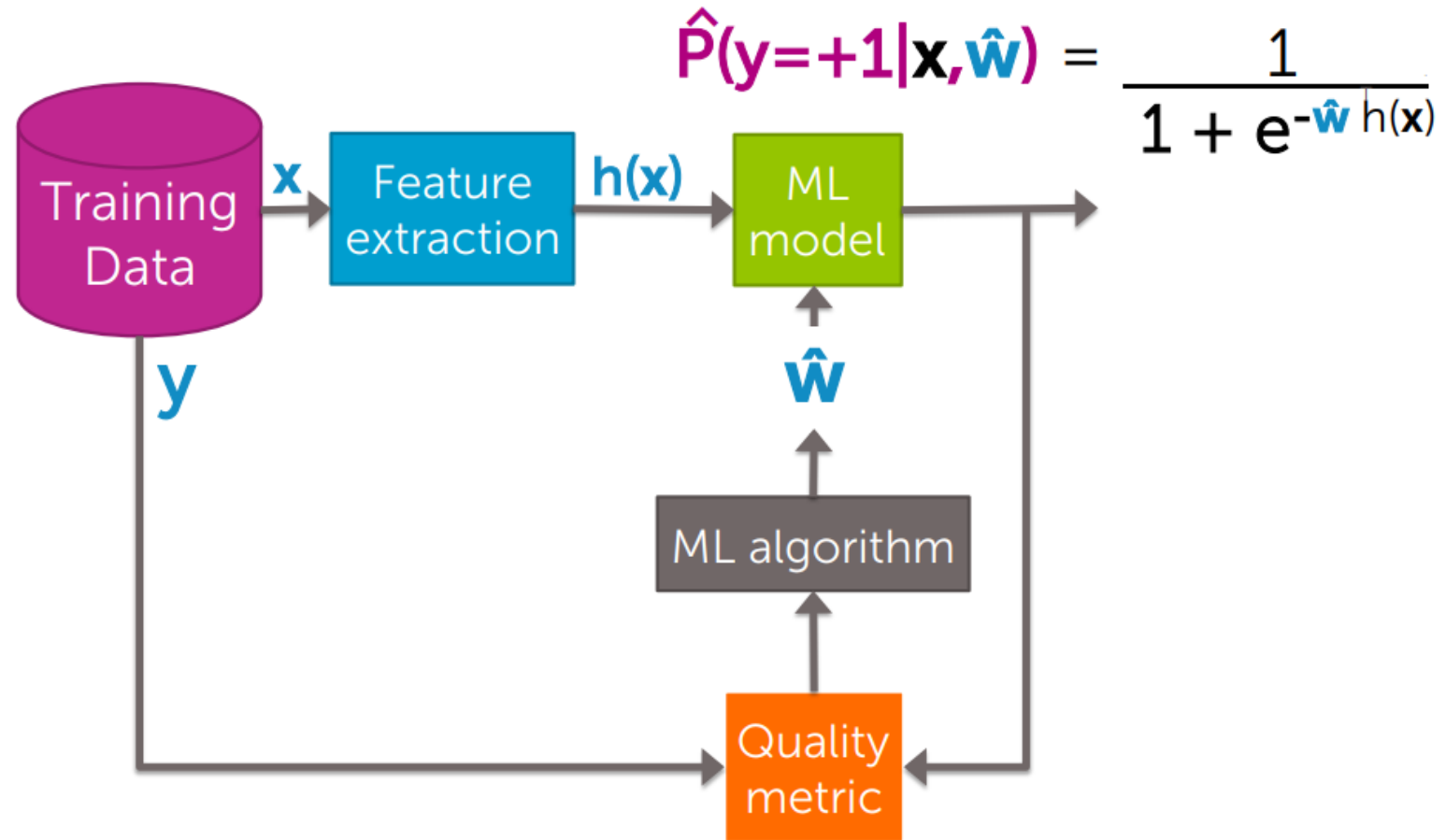
Find "best" classifier =

Maximize quality metric over all possible w_0, w_1, w_2

Likelihood $\ell(\mathbf{w})$



Modelo de regresión logística



Vamos al código...